AgentForge: Deterministic Gates for an LLM Multi-Agent Pipeline from Product Idea to Deployed Web Application

***Abstract*—Large language models can now write most of a web application, yet the agent's own report that the work is finished remains the weakest evidence in the process. AgentForge is a desktop system in which four LLM agent roles (requirements and design, prototype and build, quality assurance, and deployment) take a product idea through an adaptive interview, an approved plan, a requirements specification, wireframes, a prototype, a phased build, browser-level testing and a cloud deployment. The roles share one tool-calling engine and hand off through a project record on disk, and every hand-off is checked by code: a depth floor on the plan, coverage and evidence audits on the specification, a named-page gate on the build, requirement verdicts computed from test output, and a probe of the deployed HTTPS address. The user approves each major stage and can change any stage by chat or by pointing at the page, and typed personal data is replaced with placeholders before every model request. During development, writing the specification as five parallel parts cut its generation for a 31-page store from 546 s to 141 s; a build conversation that grew to 900,000 tokens made each call about four times slower, which moved the hand-off to files; and computed requirement status corrected a report of 0 of 116 requirements met after every test had passed. A controlled evaluation protocol is specified.**

***Index Terms*—autonomous software engineering, large language models, multi-agent systems, requirements engineering, software testing, deployment automation, data privacy**

I. Introduction

Ask one model to build a thirty-page web application in a single conversation and it tends to return a smaller one. In AgentForge's early runs the model said as much: partway through, it decided it "must be honest about capacity" and finished a coherent subset of the approved pages. No file it wrote was wrong. The application was simply smaller than the one the user had approved, and only a count against the plan could show by how much.

Code-trained language models have brought function- and module-level programming within reach of automation [1], [2], and agentic systems now resolve a measurable share of real repository issues end to end [3], [4]. The hard part has moved. It is the distance between code that runs and an application someone can trust: the requirements were understood, every promised screen exists, tests show the behaviour the user asked for, and the deployed address answers. Surveys of LLM-based multi-agent systems for software engineering name quality assurance and evaluation as the open problems [5].

AgentForge is built on one rule: a model may write anything, but code decides whether a stage is finished. Four agent roles take turns on one tool-calling engine. They pass work to each other through files in the project's own record, and every stage boundary has a check that reads those files and either lets the project through or names what is missing. The model is used where judgement is needed (choosing the next interview question, writing a page, fixing a failing journey) and kept out of the decision about whether that work is done. The paper makes four contributions:

1) A pipeline from a one-sentence idea to a live HTTPS address in which four agent roles share one engine and one on-disk project record, and the user approves and can steer every stage (Section III).
2) Deterministic checks at each hand-off, including requirement status computed from test-runner output rather than from the agent's report (Section III-E).
3) A privacy layer that keeps typed personal data out of model requests while the artifacts keep the real values (Section III-G).
4) Measurements of where a long-running agent pipeline broke and what fixed each break, with a protocol for a controlled evaluation (Section V).

II. Related Work

A. Code Generation and Repository Agents

HumanEval-style benchmarks measure whether a model can turn a docstring into a working function [1]; SWE-bench asks whether an agent can resolve a real GitHub issue [3]. Open-weight families such as Qwen2.5-Coder [2] and Gemma 3 [6] run on a single workstation, which matters for a tool that must work with a locally hosted model. ReAct set the pattern of interleaved reasoning and tool calls that most coding agents follow [7]. SWE-agent showed that the interface an agent is given changes how often it succeeds [4], and Agentless reached results competitive with open-source agents on SWE-bench Lite with a fixed localise, repair and validate pipeline [8]. AgentForge takes the second lesson further: the stage order and every stop condition are fixed in code, and the agent loop runs only inside a stage.

B. Multi-Agent Software Engineering

LLM-based multi-agent systems differ mainly in how agents coordinate. ChatDev runs a virtual software company through chat-chained waterfall phases [9]. MetaGPT encodes standard operating procedures as roles that exchange structured documents [10]. AutoGen frames collaboration as programmable multi-agent conversation [11], and AgentCoder separates a programmer agent from a test designer and a test executor, showing that independently written tests improve the final code [12]. As described in their papers, these systems start from a requirement given as text and end with source code. Interviewing the user, holding the result to a plan the user approved and carrying the application to a deployed address are outside their scope (Table I).

TABLE I. CAPABILITIES AS DESCRIBED IN EACH SYSTEM'S PUBLICATION

Capability	ChatDev [9]	MetaGPT [10]	AutoGen [11]	Agent-Coder [12]	Agent-Forge
Role-specialised LLM agents	Yes	Yes	Yes	Yes	Yes
Interviews the user for requirements	No	No	No	No	Yes
Specification checked by code against an approved plan	No	Partial	No	No	Yes
Progress decided by executed tests	Partial	Partial	Partial	Yes	Yes
Browser journeys and screenshot review	No	No	No	No	Yes
Deployment verified at a public address	No	No	No	No	Yes
Personal data hidden from the model	No	No	No	No	Yes

Partial: MetaGPT's structured documents are not checked against a plan the user approved; ChatDev, MetaGPT and AutoGen feed run errors back without a pass criterion fixed in advance.

C. Self-Checking and Long Contexts

Self-Refine loops a model over its own critique [13], and Reflexion turns failure feedback into verbal memory across attempts [14]; both let the model judge its own work. Without external feedback, models do not reliably correct their reasoning [15], and LLM judges show position, verbosity and self-enhancement biases [16]. Models also use the middle of a long context less well than its ends [17]. AgentForge keeps model critique where a picture or a document must be read, but pass or fail is always a function of files on disk.

D. Requirements Engineering

ISO/IEC/IEEE 29148 lists the properties of a well-formed requirement, among them being necessary, unambiguous, singular and verifiable [18]. Public corpora such as PURE [19] make requirement quality measurable, and multi-agent LLM pipelines have been used to derive UML models from natural-language requirements [20]. AgentForge checks a subset of these properties mechanically and draws its diagrams from the specification it has already checked (Section III-C).

III. SYSTEM DESIGN

AgentForge runs on the user's computer. A Studio interface talks to a Python backend, the backend keeps one engine per project, and the engine runs whichever agent role the current stage needs. Fig. 1 shows the parts and the checks between stages.

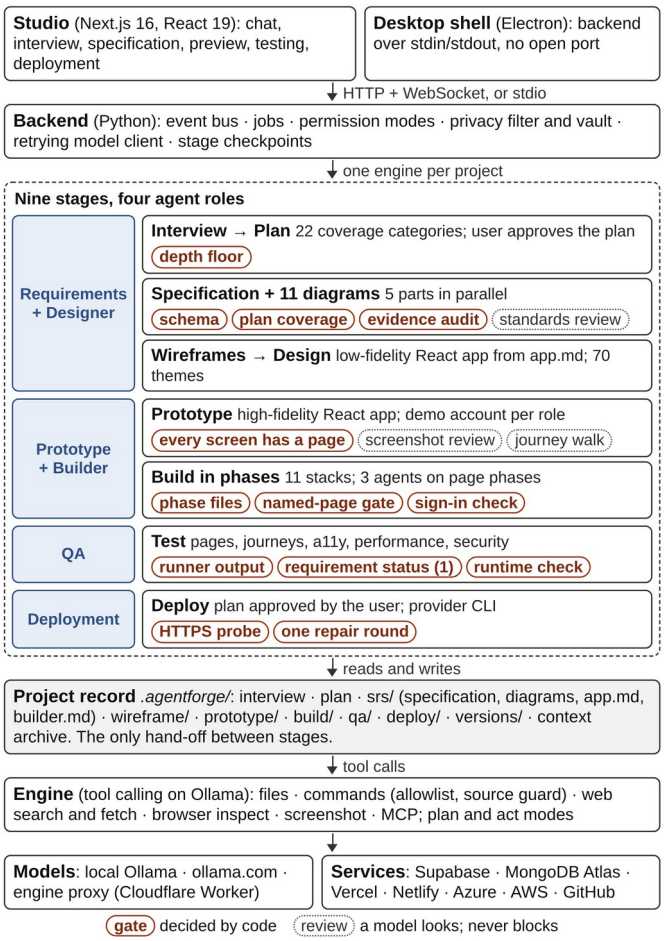


Fig. 1. AgentForge architecture. Four agent roles share one engine and hand off through the project record; the checks between stages are decided by code.

A. Architecture

The pipeline has nine stages: interview, plan, specification, wireframes, design, prototype, build, test and deploy. Four agent roles own them (Fig. 1). A role is not a separate process or a separate model. It is a set of prompts, a stage of the project and a label on the events the Studio shows. All four run on the same engine against the same workspace, and every instruction a model receives is a Markdown file under *prompts/*, so changing how a stage behaves means editing a file, not the code.

Each project has a record folder, *.agentforge/*, beside the application it builds. The interview, the plan, the specification, the wireframe and prototype apps, the build's phase state, the test runs and the deployment record live there. A later stage reads what an earlier one wrote, the Studio's views read the same files, and a stopped stage resumes from its saved state.

B. Engine and Context

The engine is a terminal coding agent built on Ollama's tool-calling API [21]. It exposes ten tools: list, read and search files; write a file or replace one exact span; run a command; search and fetch the web; inspect a local page in a headless browser; and take a screenshot that a vision-capable model is then shown. More tools can be attached through the Model Context Protocol [22]. Any Ollama model with tool calling can drive it, locally or on ollama.com.

A stage runs in two modes. In plan mode the write and command tools are withheld and the model produces a numbered plan. In act mode it carries the plan out and must end with an explicit marker: complete or blocked, and, during an

audit, verified or incomplete. The runtime bounds the loop at eight execution rounds, three audit rounds and three consecutive rounds without a tool call, after which the stage is declared blocked. When a step needs a JSON answer, code validates it and sends the validator's message back in the same conversation, up to three times.

The first design kept one conversation per project and compacted it into a summary past 88% of the model's window. Section V-C shows why that was not enough. Each stage and each build phase now starts from a small conversation; earlier turns are archived verbatim and the inputs are read from the record.

C. Requirements: Interview, Plan and Specification

The interview asks one question at a time. The first, which kind of product this is, comes from a catalogue of ten product types and costs no model call. Every later question is chosen by the model from what is still unknown across 22 requirement categories (purpose, roles, workflows, data, permissions, business rules and so on), each tracked as known, partial, unknown or not applicable. A one-screen tool finishes in a handful of questions and a multi-role system takes more, up to a ceiling of 25, after which the model writes a summary for the user to confirm or correct. A user who skips the interview gets a plan that states every assumption it made.

The plan is what the user approves, and it is a ceiling: the specification's pages come from its screens, its tables from its records and its roles from its users. Its depth floor requires a product intent of at least 120 characters and a purpose for every screen. In a record-based application every record keeps at least two fields and at least one workflow has two or more steps. If the product has sign-in, at least one role needs two things it can do, the plan must say how accounts are created, and open sign-up may not create a privileged role such as an administrator, manager or cashier.

The specification is written in five parts at once (frame, tables, API, requirements and quality) from page routes and table names computed once from the plan, then joined and validated against a Pydantic schema that requires at least three functional and three non-functional requirements. The *coverage gate* then copies in what the plan states and the draft omitted, and counts what is still missing: a planned screen with no page (routes compared without letter case or parameter names), a record with no table, a feature, workflow or role ability with no requirement, and a functional requirement with no traceability row. Any remaining gap stops the stage.

Two quality checks follow. The *evidence audit* is seven fixed rules: both requirement sets exist; identifiers follow FR-### or NFR-### and are unique; each requirement says "shall"; each names a verification method from a fixed list; each functional requirement has a rationale; each non-functional requirement carries a number with a unit or a named standard; and each functional requirement has exactly one traceability row with a test case. A model reviewer then scores functional, non-functional, security, ambiguity and traceability quality from 0 to 5 and lists findings. A finding that cites a requirement not in the document is rejected, every failed audit rule is added as a blocker, and Python, not the reviewer's verdict, decides that the draft passed: no blocker and every score at least 3. Repairs are JSON edits applied only if the result still validates. By default one review round runs and its result is stamped on the document.

Eleven diagram kinds are then drawn in Mermaid, each in its own call and six at a time: system context, use case, entity-relationship, sequence, activity, class, state machine, data flow, BPMN, component and deployment. Each is rendered to SVG, photographed and reviewed by a vision model for clipped labels, crossing lines and unreadable scale, for at most two rounds; a diagram that comes back worse is restored. Two hand-off files close the stage. Both are projections of the specification made by code, since a document a model writes can be truncated. *app.md* holds what the user asked for, in their words, and the site map; *builder.md* holds the full specification for the build, the tests and the deployment.

D. Wireframes, Design and Prototype

The wireframes and the prototype are both small React applications (Vite, TypeScript, Tailwind CSS and shadcn/ui) that the agent writes with its own file tools, following Anthropic's web-artifacts-builder skill [23]. The Studio creates each app and bundles it into one self-contained HTML page; the pages are the agent's. Wireframes are drawn from *app.md* alone, so they show what the user asked for rather than what the specification inferred. For ten or more screens, one agent writes the shared layout and the first three pages and three agents write the rest in parallel.

The user then picks one of 70 design themes, each a DESIGN.md file the prototype must follow. The prototype is a new high-fidelity app written from the approved wireframes, the theme and *app.md*. Where the product has sign-in, code derives one fictitious account per role from the specification; the prototype signs in with these and the build later seeds the same ones. Two reviews follow. Every page is photographed at 1440 and 390 pixels wide and a vision model lists visible defects, up to 40 of which are fixed. Each user journey (up to 24, of up to ten steps) is then walked in a real browser: the walker signs in as the journey's role, reaches each screen by clicking links the pages actually have, asks the model what to press or fill, and photographs every step. These are reviews, not gates, and are skipped when the model cannot see pictures.

E. Build and Verification

The builder installs one of eleven stack templates (Next.js, Vite or Remix on Supabase or MongoDB, and a MERN microservice layout, with or without a Supabase Storage bucket for uploads) into an empty workspace, never over existing work. A preflight step checks Node.js, a browser and the stack's database before anything is planned, and lists everything missing in one prompt. The model may then ask the user up to five questions it cannot settle itself. After that the build never stops to ask. A decision with no safe default takes the most conservative reading of the specification, a missing credential becomes real code against a documented environment variable, and each such choice is recorded in the build report as a gap labelled *gap*, *unavailable*, *untested* or *known*.

The Studio, not the model, holds the list of phases (Fig. 2): a foundation (data models, authentication, shared layout), page phases of four approved pages each, a wiring phase with the seed data, and the unit and journey tests. Once the foundation stands, three agents build page phases in parallel and the two

test phases run together. Each phase starts a fresh conversation that begins with the build request. A phase is finished when its progress record names files that exist and are not empty; otherwise it is asked for again, at most twice. When every phase has run, an approved page that is still not built fails the build by name. A sign-in page that displays demo credentials is rejected as well.

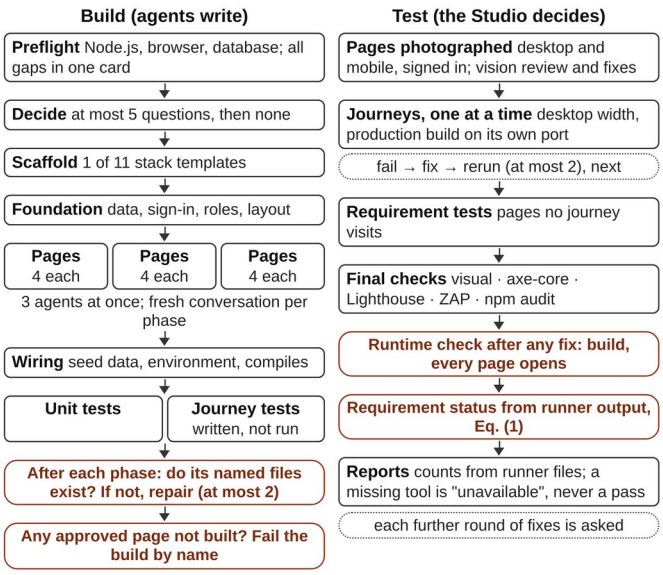


Fig. 2. Build and test sequence. Agents write the application and fix what fails; the Studio decides when a phase, the build and each requirement are done.

Testing follows a fixed order. Every page is photographed whole at desktop and mobile width, signed in as a role that may open it, and reviewed for visible defects. The Studio then runs each user journey on its own with Playwright [24] against the production build, on MongoDB stacks with a fresh test database that is dropped afterwards. A failing journey is fixed and rerun, at most twice, before the next one runs, and pages that no journey visits get requirement tests of their own. The agent then runs the remaining layers once each: visual comparison, accessibility with axe-core [25], performance with Lighthouse [26], a ZAP baseline scan [27] and a dependency audit; a fix made then is checked by building the app and opening every page. The agent may not run the whole journey suite, which the Studio runs once and owns. A missing tool is recorded as unavailable, never as a pass, and every count is read from the runners' own output files.

Requirement status is computed, not reported. Let *P(r)* be the pages that the traceability row of requirement *r* names, *J(p)* the journeys with a step on page *p*, and *pass(j)* true when the desktop test of journey *j* passed. When no test names *r* in its title,

$$\mathrm{met}(r) \;\Leftrightarrow\; P(r) \neq \varnothing \;\wedge\; \forall p \in P(r)\;\; \exists j \in J(p) : \mathrm{pass}(j) \qquad (1)$$

A requirement is *failed* when some page in *P(r)* is visited only by failing journeys and *untested* otherwise. When tests do name *r*, they decide on their own. The result overwrites whatever the builder wrote and is marked as decided by the Studio.

F. Deployment

Deployment is planned first, like a change request (Section III-H). The targets are Vercel, Netlify, Azure App Service, AWS EC2, AWS ECS and GitHub, filtered by what the stack supports. The agent works through each provider's own command-line tool, signed in through that tool's browser or device-code login, so no token is pasted. Its rules favour one pass over repetition: no test is rerun, the application is built once, at most three questions are asked, and the smoke proof is at most six read-only requests. When the agent records the run as live, the Studio checks the result itself. It requests the public address and every check the agent recorded, and rejects an address that is not HTTPS or whose host is local or a private IP, and any status outside the expected range; 429 and gateway errors from a host that is still starting are retried after 5 s and 10 s. A failure goes back to the agent once. If the address still does not hold, the run ends as failed, with the cause.

G. Safety, Privacy and Reliability

Every command the agent runs is checked against an allowlist of 132 program names and a deny list (*sudo*, a download piped into a shell, Invoke-Expression, deleting the root or home directory, a credential file sent to a network command). The code calls this defence in depth, not a sandbox. AgentForge's own source is snapshotted before every command and restored if the command changed it.

Personal data does not reach a model in clear text. Everything the user types passes through Microsoft Presidio [28] with a spaCy English model, set to 15 entity types including Sri Lankan identity and phone numbers. Each value found is kept, encrypted, in a per-project vault on the user's computer under a placeholder numbered for the whole project (<PERSON_1> is the same person in every message). The one client every model call passes through hides known values on the way out and restores them in the answer and in the arguments of tools that act on the local machine; web search, web fetch and plugin tools keep the placeholders. If Presidio is missing, the message is not sent. Text shaped like a key or a connection string is refused in the chat and must be saved in Settings instead.

A transient model-service failure (a reset connection, a timeout, HTTP 408, 425, 429 or a 5xx error) is retried up to ten times with back-off from 2 s to 30 s. Every stage writes a checkpoint when it starts, finishes or fails, so a run cut off by a restart resumes. Hosted models are reached with the user's own ollama.com key or through a Cloudflare Worker that holds the service key, passes through eight API routes and meters calls against the user's account.

H. Human in the Loop

The agents do the work, but the user decides what is built and when a stage moves on. Nothing passes the plan, the specification, the design or a deployment without the user's approval, and every stage has a chat box and an artifact the user can change (Table II). Questions are rationed to what only the user can decide: at most 25 in the interview, five before a build, three per change request and three per deployment. Everything else is a stated default that the user can correct.

TABLE II. WHAT THE USER CAN DO AT EACH STAGE

Stage	User actions
Interview	Pick an answer or type one; attach screenshots and documents; correct the summary; skip
Plan	Approve, or revise in their own words; each revision is a new version
Specification	Approve; ask for a revision; answer each open decision (the first option keeps the assumption)

Stage	User actions
Wireframes	Ask in chat; select, draw on or retype any part of a page
Design	Choose 1 of 70 themes or customise: 32 palettes, 117 heading and body fonts, own pictures, a written brief
Prototype	Ask in chat; select, draw on or retype; approve before the build
Build	Connect missing accounts; answer up to five questions; edit code; allow or stop further fix rounds
Test	Choose whether another round of fixes runs; ask for failing journeys to be fixed
Deploy	Approve, revise or cancel the plan; answer up to three questions; sign in to each provider
After build	Request changes in chat, planned first; select or draw on the running app

A message typed at any stage is answered from the project or applied to it, and the agent updates every artifact the request reaches (the specification, the wireframes, the prototype or the build) without widening it. Once a specification exists, a change is planned first: the planner reads the project, may ask up to three questions and shows a plan to approve, revise in the user's own words or cancel. Nothing is touched before approval, and the plan step can be switched off per project. Each approved change becomes a version (v1, v2, ...) that lists the files it touched and the test results before and after.

Users can also point. On the wireframes, the prototype and the running application, a select tool picks an element and a pencil marks an area; the Studio describes what was picked or drawn round (tag, words, attributes, place on the page and a slice of markup) and attaches a screenshot for the agent. A text tool changes words in place: when they occur once in the source, the Studio swaps them and rebundles the app with no model call; otherwise it hands the change to the agent. Code can be edited directly in the Studio's code pane.

The user also controls a run while it happens. The permission mode decides what the agent may do unasked, and a permission card offers allow, allow without asking again, or deny with a note for the agent. Stop ends a model request between tokens and a running command with its process tree. Further rounds of fixes are a Yes or No card, and a typed yes in English or Sinhala counts. The model and its reasoning effort can be changed per project without losing the project's record.

IV. Implementation

AgentForge Studio 1.0.3 has about 32,300 lines of Python in the backend, the engine and the agents, 27,800 lines of JavaScript in the Studio, and 19,500 lines of Markdown prompts, including the 70 design themes and the deployment skill pages. Table III lists the technology at each layer. Three choices shape the rest. The backend has no web framework; it runs on the Python standard library, so the desktop build carries an embedded interpreter and seven Python packages. Every model call goes through one client, which is where retries, account metering and the privacy filter live. And every artifact is a file in the project record, so any stage can be stopped, inspected and resumed.

TABLE III. Technology Stack

Layer	Technologies	Used for
Studio	Next.js 16, React 19, Tailwind CSS 4, Zustand 5, Lucide, JSZip	Chat, stage views, previews, testing and deployment panels
Desktop	Electron 44, electron-builder 26, agentforge:// scheme, C# online installer	One window; backend over stdio
Backend	Python 3.10+ standard library (ThreadingHTTPServer, RFC 6455 feed, threads), Pydantic 2, httpx, cryptography (Fernet)	API, jobs, event bus, schema, vault
Engine	Ollama Python SDK, ten tools, MCP client (JSON-RPC 2.0 over stdio)	Tool-calling loop of every agent role
Models	Local Ollama; ollama.com; Cloudflare Worker proxy with accounts	User-chosen model; vision detected per model
Privacy	Microsoft Presidio, spaCy en_core_web_sm	15 entity types hidden from models
Documents	Mermaid CLI; marked and Puppeteer	SVG diagrams; specification PDF
Wireframe and prototype kit	Vite 5, React 18, TypeScript 5, Tailwind 3.4, shadcn/ui (Radix), Recharts, vite-plugin-singlefile	One self-contained bundle.html per app
Browser	Playwright Chromium; Edge or Chrome in headless mode	Screenshots, page inspection, journey walk
Generated apps	Next.js 15; Remix 2 and Vite 6; Express 4; Supabase (supabase-js 2, @supabase/ssr); MongoDB with Mongoose 8; bcryptjs, jose	Eleven stack templates
Generated tests	Vitest, Testing Library, Supertest, Playwright 1.62, axe-core, Lighthouse CI, ZAP, npm audit, GitHub Actions	Unit, journey, visual, a11y, performance, security
Cloud	Supabase OAuth; MongoDB Atlas CLI; gh, Vercel, Netlify, Azure and AWS CLIs	Provisioning, sign-in, deployment
Integrations	20 providers: payments, e-mail and SMS, image storage, social sign-in, maps, analytics	Optional features, keys encrypted

A. Backend Runtime

The API is a ThreadingHTTPServer bound to 127.0.0.1:7824, and the live feed is an RFC 6455 WebSocket server on port 7825 written on the standard library; the Studio proxies */__agentforge/api* and */__agentforge/ws* to them. In the desktop app nothing listens: the window and the backend exchange one JSON object per line over standard input and output, and the event feed travels the same way. Work that needs a model never runs inside a request. It becomes a background job on its own thread, and the browser polls for the result, so a specification that takes ten minutes holds no connection open. Every event a stage emits goes to an in-memory bus and is appended to *events.jsonl*, so a browser that reconnects replays the same conversation. The project's conversation is written to *context.json* after every turn, turns removed by compaction are appended verbatim to *context-archive.jsonl*, and a checkpoint at the start, end or failure of every stage lets an interrupted run resume. A project only moves forward through the stage order.

B. Engine Logic

One request to the engine is a loop of at most 40 model turns. Each turn sends the conversation with the tool schemas (without the write and command tools in plan mode), executes the returned tool calls and appends their results. Calls that write different files are run up to four at a time, which the wireframe and prototype stages use; edits and commands stay in order. A turn ends the loop when it carries no tool call or a completion marker; a plain closing sentence such as "the task is complete" is accepted when a model omits the marker, unless it also says the work is incomplete.

Context is estimated at four characters per token plus 1,500 per picture. Past 88% of the window, older turns are cut into stretches of up to 100,000 characters, summarised four at a time and merged in groups until one memory remains; a stretch the model will not summarise becomes a mechanical digest of the requests, commands and file paths it contains. When the service rejects a request as too long, the token count it reports rescales the estimate and the history is compacted again, up to three times. A screenshot is attached only while the whole request stays under 2.4 MB, and older pictures are replaced by a note. Structured answers go through a validator, whose error is sent back for at most three attempts, and repairs to the specification are JSON edits (*set*, *add*, *remove*) addressed by path and item key, such as *functional_requirements[id=FR-012]*.

Independent work runs in parallel lanes: five specification parts at once, six diagrams, four compaction workers, and three agents for wireframe pages, page phases and test phases, each agent in a conversation of its own that is archived when it ends. The interview's bookkeeping of facts and coverage runs on a background thread while the user reads the next question. Algorithm 1 shows how the build combines these parts.

Algorithm 1 Phased build with verified phases

Input: approved routes R, stack s, the user's direction d, build request q
$P \leftarrow$ [foundation] + chunks(R, 4) + [wiring, unit tests, journey tests]
$f \leftarrow$ SHA-256(R, s, d); if a saved state has fingerprint f, keep its done phases
for each phase $p \in P$ not done **do**
  if p starts a run of independent phases **then** build the run in 3 parallel lanes
  else build p in a fresh conversation that begins with q
  repeat at most twice **while** problems(p) $\neq \varnothing$: repair(p, problems(p))
  mark p done if problems(p) $= \varnothing$, else incomplete; save state
if some $r \in R$ has no recorded, non-empty file **then fail** naming r
review pages; run journeys one by one; run final checks; set requirement status by Eq. (1)

C. Verification Logic

A page phase is checked against its own progress record, *build/progress/<phase>.json*, which must name files that exist inside the workspace and are not empty. The Studio's journey run is keyed by a SHA-256 fingerprint over the path and content of every source file, skipping dependencies, build output, test output and the record: while the fingerprint is unchanged, a later Testing run reuses the result instead of running again. The runner script builds the app once, starts the production server on a free port, runs Playwright with one retry and two workers, and stops only the server it started; on MongoDB stacks it creates a database named within Atlas's 38-character limit and ending in *_e2e*, and drops it afterwards. The suite has 45 minutes, the page photographs 20 and the runtime check 10, and screenshots are copied out as the tests take them, because the next run empties Playwright's output folder.

The specification checks are plain functions. Routes are compared by a key that ignores letter case and parameter names, so */Admin/[branchId]* and */admin/:id* are one page; record names are matched to tables through singular and plural forms. The evidence audit scores 100 times the share of its seven rules that pass, capped at 50 when either requirement set is missing, and any score below 100 adds blocking findings. The deployment planner must have opened the target's skill pages with *read_file* before its plan is accepted, and the live probe accepts a status range written as 200, 2xx or 200–399.

D. Safety, Privacy and Model Access

A command is split on &&, ||, ; and | outside quoted strings, and the program of each part is checked against the allowlist; PowerShell cmdlets pass by their Verb-Noun shape except six that run code or another process. A command may run 600 s, a deployment release 1,800 s, and a server start 75 s; on Windows every process starts without a window. Text search skips dependencies, build and test output and binary files and stops after 20 s. Before each command the files of AgentForge itself are snapshotted, and afterwards any change is undone unless the repository's HEAD moved, which marks a deliberate pull or checkout.

Presidio analyses typed text in 50,000-character chunks and keeps matches scoring at least 0.4, with added recognisers for Sri Lankan identity numbers and phone numbers in ten regions. Values go to a Fernet-encrypted vault under placeholders such as <PERSON_1>. The model client replaces known values in every message, tool result and earlier tool call before sending, and restores them in the reply; in a streamed reply it holds back a chunk that ends with what may be the start of a placeholder until the next chunk completes it. Provider keys for the 20 optional integrations are encrypted the same way.

The client retries HTTP 408, 425, 429, 500, 502, 503, 504, 522 and 524 and broken transports up to ten times, waiting 2, 4, 8, 15, 20 and then 30 s, and Stop is checked every 0.1 s, so a run never waits for a reply it no longer wants. Replies from hosted models are streamed, since a single response is cut off by the service's edge after about 100 s without a byte. Whether a model can see pictures is read from its own capability list and cached for a week. The Cloudflare Worker compares the app token in constant time, rate-limits each caller, passes through eight routes, and doubles as the OAuth broker for Supabase sign-in, so its client secret never ships with the app. Each model call is counted as one request against the limit an administrator sets for the account.

E. Distribution and Tests

The release script exports the Studio as static files, stages the backend with its Node modules and an embedded Python 3.12, packages Electron, pins the command-line tools, and compiles a 150 KB online installer with the C# compiler built into Windows; the installer checks every file against a SHA-256 manifest. The desktop app serves the Studio from an *agentforge://* scheme, and every process it starts is its child, so quitting ends them all. Models are chosen by the user: the bakery project in Section V-B used gpt-oss:120b through Ollama's cloud, and accounts default to DeepSeek V4.1 Flash. The repository holds 1,733 Python unit tests and twelve Node test files, which cover the engine, every gate, the privacy filter and the deployment flow with stand-in models instead of live calls.

V. Evaluation and Preliminary Results

A. Research Questions and Protocol

The observations in Section V-C were gathered while the system was being built, so they cannot attribute an improvement to one component. The controlled study is the next step: a fixed suite of application ideas, from a one-screen

tool to stores of over thirty screens, built with the same model and settings under each condition of Table IV, three runs per condition, reported as mean and standard deviation. Requests are logged at the single model client, which gives exact token counts and makes leaked personal data countable.

TABLE IV. RESEARCH QUESTIONS, CONDITIONS AND METRICS

Research question	Condition	Primary metrics
RQ1: Do the coverage gate and audit make the specification more complete?	Gates on vs. first draft kept	Plan items with no requirement; audit score; rater-found omissions
RQ2: Does a phased build deliver every approved page?	Phased vs. one conversation	Pages built of pages approved; time; tokens
RQ3: How far do agent-reported statuses diverge from Eq. (1)?	Builder report vs. Eq. (1) vs. blind manual test	Cohen's κ with the human verdict; false "met"
RQ4: What do per-phase conversations cost or save?	Fresh per phase vs. one shared, compacted	Tokens and latency per call; total time; context defects
RQ5: How reliably does deployment reach a probed live address?	Each app to each allowed target	Success rate; time; repairs; questions asked
RQ6: Does the filter keep personal data out of requests?	Synthetic personal data; filter on vs. off	Values in logged requests; wrongly hidden terms

B. Development Corpus

The observations come from the authors' own projects: a bakery ordering application (interview to prototype), a 28-page application (specification to deployment), a 31-page online store and a store whose plan had 71 screens. Each row of Table V is one observed run before and, where stated, after a change, as recorded in the version history. The rows explain the current design; they are not a comparison with other systems.

C. Observations

Table V lists twelve failures in the order of the pipeline. The coverage gate gave the earliest result: on the bakery project it took the first specification draft from one diagram to seven and from three functional requirements to six. Before the changes in rows 4 to 8, the 28-page application took about seven hours of machine time from specification to deployment.

TABLE V. FAILURES OBSERVED DURING DEVELOPMENT AND THE CHANGE EACH LED TO

Failure observed	Measurement	Change made
1. One conversation builds the whole app	A coherent subset; model cited "capacity"	Numbered phases; named-page gate
2. Interview turn rewrote all facts (~4,000 tokens)	~20 s per question (up to 64 s); after: 1.9–4.6 s	Question alone; facts updated in background
3. Specification as one ~60,000-token reply	546 s, 31-page store; after: 141 s	Five parts in parallel
4. One conversation through all build phases	900,000 tokens; 21 s per call (5 s at 50,000)	Fresh conversation per phase
5. One agent draws all wireframes	About 1.5 h for 117 pages	Three agents after the shared layout
6. One agent builds all pages	2 h 37 min for 31 pages	Three agents on page phases
7. All journeys run, then fixed together	About 1.5 h, 31-page store	One journey at a time
8. Search walked test frames and packages	16 min for one search; deploy plan 37 min	Skip outputs and binaries; 20 s cap
9. Requirements marked before tests ran	0 of 116 met after all tests passed	Status from runner output, Eq. (1)
10. Routes compared with letter case	117 pages for 71 screens	Case-insensitive route keys
11. "Admin", "Require" taken for names	117-page wireframe ended with 2 pages	Names need two name-shaped words
12. Screenshot added to a long conversation	2.5 MB answered; 3.4 MB got HTTP 500	Pictures only below 2.4 MB

D. Discussion

Shared files beat a shared conversation. The first design treated the conversation as the project's memory. It failed on cost, not correctness: by the build every call resent hundreds of thousands of tokens, and a request that also carried a screenshot exceeded what the service accepted. The switch to small conversations was safe because every stage's input was already a file (the plan, the specification, the hand-off files, the per-phase progress records). This fits the evidence that models use long inputs unevenly [17].

Self-reports are the weakest evidence. Rows 1 and 9 of Table V are one failure in two stages: the model's account of its work differed from the files. Each fix replaced a statement by the model with a function of what is on disk, consistent with the finding that models do not reliably correct themselves without external feedback [15]. The specification reviewer's verdict is recorded but not obeyed for the same reason.

A check that fires wrongly does damage. With "require" taken for a name, the agent read <PERSON_3>("fs") and wrote it back into the code; one capital letter in a route produced 46 duplicate pages. A check that rewrites content needs precision more than recall, and both checks were narrowed until they fired only on the case they were written for.

VI. LIMITATIONS AND FUTURE WORK

Five limitations bound these claims. First, the evidence is internal and uncontrolled: single runs on the authors' projects with no baseline. Second, the coverage gate matches plan items to requirements by shared words, and the evidence audit checks the form of a requirement (an identifier, "shall", a unit), not its truth. Third, Eq. (1) is only as strong as the journeys: a journey that visits a page but asserts little still marks that page's requirements met. Fourth, the screenshot reviews need a vision-capable model, the screenshots it sees carry the real personal data, and name detection uses an English model. Fifth, the system targets JavaScript web applications on Supabase or MongoDB, ships for Windows, and its command checks are not isolation. Next come the controlled study, journey assertions tied to acceptance criteria, and agent commands run in a container.

VII. CONCLUSION

AgentForge started from the view that the hard part of autonomous software engineering is not writing code but knowing when it is done. It hands that decision to code at every boundary and leaves what to build, and when to move on, to the user. Building it showed two things the first design did not expect: a conversation is a poor place to keep a project, and every place where a model's report stood in for evidence eventually gave a wrong answer that a short function over the files could correct. The study of Section V-A will test whether this holds beyond the projects the system was built on.

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